Maths Teacher, 8C

Vasant Valley School, New Delhi

Subject: Maths

3139/M8C/0027

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Maths Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Maths are:

Integers and rational numbers, powers and bases
Applications in daily life: word/visual problems

Great job nurturing these talents!

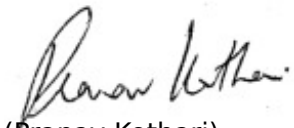
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Maths :

Mensuration: area and perimeter; volume
Algebra: concepts and applications

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student↓	Question No: → / Correct Answers →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
			C	D	A	B	A	C	C	D	C	A	A	B	D	A	C	C	D	D	C	D	C	B	C	D	D	B	C	D	D	C	C	B	B	D	C	C	D	C	D	B				
1	Advait G Singh		✓	✓	✓	✓	✓	D	✓	✓	D	✓	✓	A	✓	✓	D	✓	C	✓	A	✓	D	C	B	B	C	D	✓	✓	A	✓	✓	C	C	✓	B	✓	✓	✓	✓	✓	24	567	77	
2	Aiyana Kathuria		✓	✓	C	✓	✓	A	✓	B	B	✓	B	C	C	✓	D	✓	✓	✓	A	✓	B	✓	B	✓	A	✓	B	A	✓	A	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	24	560	75	
3	Alisha Ahmad		✓	C	✓	A	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	D	D	✓	B	C	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	B	✓	✓	30	658	92	
4	Anik Mallick		D	✓	✓	✓	B	B	✓	✓	D	C	C	✓	C	D	A	A	B	C	D	B	B	D	D	A	B	A	✓	B	B	*	A	C	D	A	A	✓	B	B	C	D	8	367	3	
5	Anvay Surana		✓	✓	✓	C	✓	B	D	✓	B	✓	✓	✓	C	✓	D	✓	✓	✓	✓	✓	✓	D	A	A	A	✓	✓	✓	A	A	✓	C	C	✓	✓	✓	C	A	✓	✓	24	589	82	
6	Anvita Bose		✓	C	✓	A	B	✓	D	✓	D	✓	✓	D	✓	✓	B	✓	✓	A	✓	✓	✓	C	D	A	A	A	B	✓	✓	A	✓	✓	A	C	✓	B	A	✓	B	C	19	534	68	
7	Arnavi Jain		✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	A	D	B	A	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	C	B	✓	✓	31	685	95	
8	Arzoi Singh		B	C		D	✓	✓	✓	✓	B	B	D	C	C	✓	B	✓	B	✓	✓	A	A	C	B	A	A	C	A	B	✓	B	✓	C	✓	✓	✓	✓	B	✓	✓	✓	17	487	50	
9	Ashyra Kachru		✓	C	✓	✓	✓	A	D	C	B	C	✓	A	C	✓	A	✓	C	✓	✓	A	B	C	✓	B	A	✓	✓	B	C	A	A	✓	D	C	B	✓	B	D	C	✓	15	460	37	
10	Cyra Parhawk		✓	C	B	A	C	B	✓	B	B	B	✓	✓	C	✓	✓	D	C	✓	B	✓	D	✓	✓	C	C	D	A	B	C	✓	✓	A	C	✓	✓	✓	✓	✓	C	✓	18	472	43	
11	Deepika Singh		✓	C	C	✓	✓	B	A	✓	A	B	✓	✓	A	✓	D	✓	C	✓	B	A	✓	✓	A	A	A	D	A	C	B	B	✓	C	✓	C	A	✓	C	B	C	C	14	465	40	
12	Donna K Chhatwal		B	C	✓	✓	✓	B	✓	✓	✓	B	✓	✓	C	✓	✓	✓	C	✓	✓	A	✓	✓	A	B	A	A	A	B	A	✓	✓	✓	✓	C	B	D	✓	✓	B	✓	22	524	65	
13	Fateh D Singh		✓	B	✓	D	✓	✓	✓	B	D	D	C	D	C	D	B	✓	✓	A	A	C	A	✓	B	✓	B	D	A	C	A	B	B		C	C	A	A	✓	✓	A	✓	12	395	8	
14	Ichcha		A	C		D	B	B			D	C	✓	D	C		B	B	C	A	✓	A	B	C				D	B		C	D	A	A	C	✓	D	A	✓	A	✓	✓	6	309	0	
15	Jai A Nath		A	C	D	A	B	✓	✓	✓	✓	✓	C	✓	✓	✓	B	D	C		✓	✓	D	C	A	B	C	✓	A	B	A	D	✓	C	✓	C	B	B	C	✓	C	A	14	447	30	
16	Jaya Grewal		✓	✓	✓	✓	✓	B	✓	✓	D	✓	✓	✓	C	D	✓	✓	✓	✓	✓	✓	B	C	D	B	A	C	B	C	✓	A	✓	C	✓	C	✓	B	A	D	C	✓	21	545	71	
17	Keshav Rana		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	C	36	800	99	O
18	Krishna		✓	✓	C	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	D	✓	✓	C	✓	✓	✓	D	✓	B	✓	✓	✓	✓	✓	B	✓	D	✓	✓	✓	D	✓	✓	✓	✓	31	670	93	
19	Mehreen Sharma		B	C	✓	✓	D	B	✓	A	D	B	✓	D	C	✓	A	B	C	A	D	A	B	C	✓	C	C	A	A	B	✓	D	✓	D	A	✓	D	B	✓	✓	✓	✓	13	417	16	
20	Namyaa Munjal		✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	B	A	✓	✓	✓	✓	B	✓	C	A	C	✓	✓	✓	✓	✓	✓	32	698	95	
21	Navya Modi		B	✓	✓	✓	✓	A	✓	✓	✓	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	B	A	✓	A	✓	✓	B	✓	✓	C	C	B	✓	✓	✓	B	A	26	609	86	
22	Pallavi Sharma		D	B	D	C	✓	B	A	✓	D	C	C	D	✓	C	✓	B	A	A	*	A	B	✓	B	B	C	A	A	✓	B	B	A	A	C	B	B	B	B	D	C	D	6	337	0	
23	Radhika Mathur		✓	✓	✓	✓	D	A	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	A	✓	A	✓	✓	C	✓	✓	✓	✓	C	✓	✓	✓	32	671	93	
24	Ridaan S Chopra		✓	✓	✓	C	✓	A	D	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	D		B	✓	✓	✓	B	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	31	660	92	
25	Sehaj Chopra		A	✓	✓	✓	✓	B	B	✓	D	✓	✓	✓	C	B	B	✓	✓	✓	D	A	A	C	B	✓	A	C	✓	✓	✓	D	✓	A	✓	C	✓	✓	✓	B	✓	✓	22	543	71	
26	Shauryaveer Pathania		✓	C	✓	✓	✓	B	✓	C	B	C	D	D	C	✓	B	✓	✓	A	✓	✓	A	✓	A	✓	C	A	✓	✓	✓	D	✓	C	✓	C	A	A	B	✓	✓	A	19	513	61	
27	Shrey Relan		✓	A	✓	✓	✓	✓	D	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	B	✓	✓	✓	34	712	96	
28	Siddharth Z Chand		✓	✓	✓	✓	✓	B	✓	A	✓	B	✓	✓	✓	✓	B	✓	B	✓	✓	✓	✓	C	B	C	✓	A	B	✓	✓	A	✓	✓	C	✓	✓	✓	B	✓	✓	A	26	613	86	
29	Uday Mehra		✓	✓	C	✓	✓	B	✓	✓	✓	✓	✓	✓	C	D	D	B	✓	✓	✓	✓	B	C	A	✓	A	✓	✓	B	✓	B	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	27	580	80	
30	Vania Taneja		✓	C	✓	✓	B	D	D	✓	B	D	✓	✓	✓	✓	D	✓	✓	✓	A	A	✓	C	D	B	A	C	✓	✓	✓	A	✓	A	C	B	B	✓	C	✓	C	A	18	519	63	

% answered correct (section)	68	53	62	71	74	32	65	74	35	50	76	59	50	76	29	79	56	68	65	62	47	35	29	21	18	44	50	56	62	26	85	29	50	56	53	65	44	65	59	65	547.1/500			
% answered correct (school students)	78	58	63	75	71	38	66	69	32	58	76	73	38	80	48	86	57	71	69	61	53	31	34	34	22	44	49	60	58	28	89	36	50	65	56	70	44	59	53	57	561.1/500			
% answered correct (all students)	68	59	53	64	70	32	50	53	33	36	64	54	31	72	38	71	40	58	58	40	39	27	32	31	23	39	29	48	47	24	74	37	39	55	44	56	32	59	48	45	502.2/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

No.	Student↓	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	**C**	**D**	**A**	**B**	**A**	**C**	**C**	**D**	**C**	**A**	**A**	**B**	**D**	**A**	**C**	**C**	**D**	**D**	**C**	**D**	**C**	**B**	**C**	**D**	**D**	**B**	**C**	**D**	**D**	**C**	**C**	**B**	**B**	**D**	**C**	**C**	**D**	**C**	**D**	**B**				
31	Vedang S Sasturkar		✓	C	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	A	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	34	744	98	
32	Vishal Sharma			✓	✓	✓	D			✓	D	B	D	✓	✓	✓	D	✓	B	C	A	A	A	✓	A	B	C	D	A	A	✓	✓	✓	C	✓	✓	A	✓	C	B	✓	D	16	476	45	
33	Vivek Kumar		B	✓	C	✓	✓	B	✓	✓	A	D	✓	C	✓	✓	B	✓	C	✓	B	✓	✓	C	B	A	A	C	✓	✓	*	*	✓	A	C	✓	B	B	A	✓	C	✓	18	504	57	
34	Ziya Khanna		✓	✓	C	✓	✓	B	D	B	B	C	✓	C	C	B	D	✓	✓	A	✓	C	D	D	✓	C	A	✓	B	✓	✓	D	✓	C	A	C	B	A	✓	✓	B	D	15	471	43	

% answered correct (section)	68	53	62	71	74	32	65	74	35	50	76	59	50	76	29	79	56	68	65	62	47	35	29	21	18	44	50	56	62	26	85	29	50	56	53	65	44	65	59	65	547.1/500			
% answered correct (school students)	78	58	63	75	71	38	66	69	32	58	76	73	38	80	48	86	57	71	69	61	53	31	34	34	22	44	49	60	58	28	89	36	50	65	56	70	44	59	53	57	561.1/500			
% answered correct (all students)	68	59	53	64	70	32	50	53	33	36	64	54	31	72	38	71	40	58	58	40	39	27	32	31	23	39	29	48	47	24	74	37	39	55	44	56	32	59	48	45	502.2/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 8 C, has performed better than the national average in the Maths ASSET paper this year.

Class 8	Maths	No. of students
8A	117%	31
8B	111%	31
8C (Your Class)	109%	34
All sections combined	112%	96

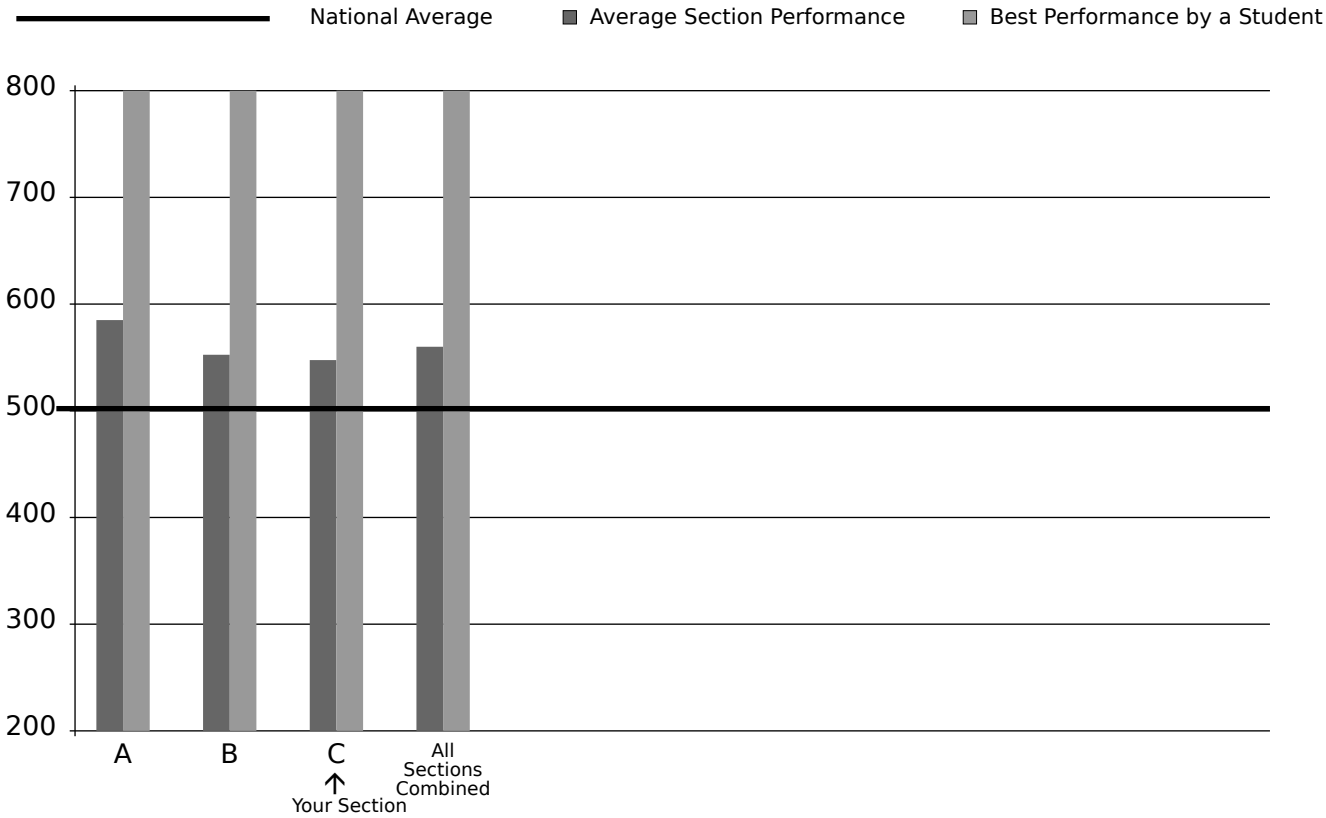
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

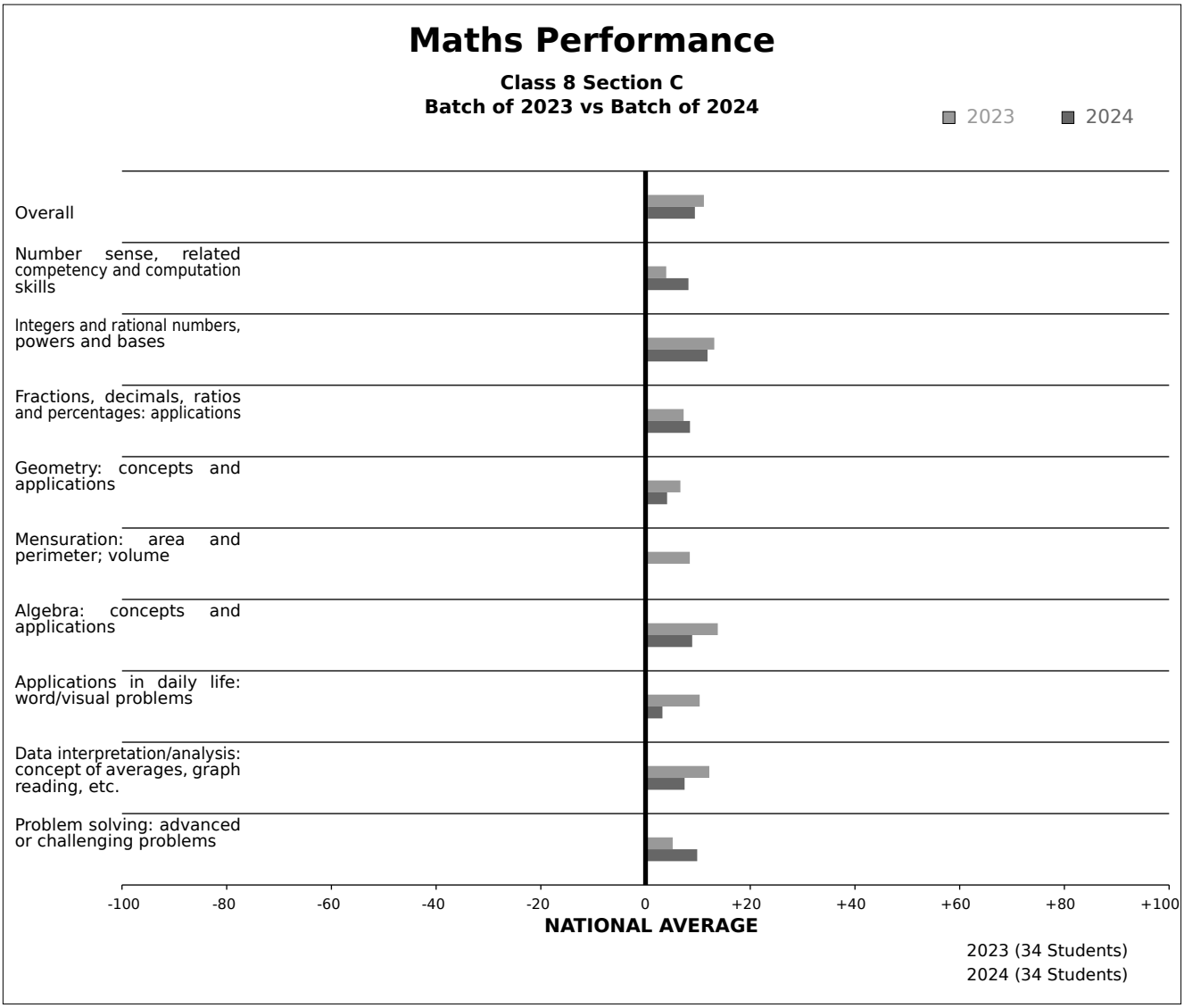
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 8 C Subject: Maths Number of Students: 34

S.No.	Skill	Questions	Class 8 C Performance	Class 8 Performance (School)	Class 8 Performance (National)
			Average	Average	Average
1*	Number sense, related competency and computation skills	3,8,19,23	57.4%	58.6%	49.2%
2*	Integers and rational numbers, powers and bases	13,14,16,20,35,36	64.2%	65.3%	52.4%
3*	Fractions, decimals, ratios and percentages: applications	11,18,21,25,29,39	54.9%	55.6%	46.4%
4*	Geometry: concepts and applications	1,9,28,38	55.9%	57.6%	51.8%
5*	Mensuration: area and perimeter; volume	4,10,15,24	42.6%	53.9%	42.3%
6*	Algebra: concepts and applications	6,17,22,33	43.4%	44.0%	34.5%
7*	Applications in daily life: word/visual problems	2,5,7,34	61.8%	64.8%	58.5%
8*	Data interpretation/analysis: concept of averages, graph reading, etc.	12,27,31,32	55.9%	61.7%	48.4%
9*	Problem solving: advanced or challenging problems	26,30,37,40	44.9%	43.2%	35.0%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Integers and rational numbers, powers and bases	1.	Mensuration: area and perimeter; volume
2.	Applications in daily life: word/visual problems	2.	Algebra: concepts and applications

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 24 Urooj cycled along a straight road for 4400 m. The diameter of her bicycle's tyres was 70 cm.

How many rotations were made by her bicycle tyres?

(Note: Take π as $\frac{22}{7}$)

A 10 **B** 20 **C** 1000 **D** 2000

Skill: Mensuration: area and perimeter; volume

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
24	28224	D	21%	31%	Your School / All Schools

Q 15 The perimeter of the parallelogram PQRS shown below is 24 cm. PS = 5 cm, ST = 4 cm and ST ⊥ PQ.

(Note: The figure is not to scale.)

What is the AREA of the parallelogram?

A 14 cm^2 **B** 20 cm^2 **C** 28 cm^2 **D** 56 cm^2

Skill: Mensuration: area and perimeter; volume

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
15	28224	C	29%	38%	Your School / All Schools

Q 32 In a parent-teacher meeting the class teacher announces - "The ratio of the number of item(s) taken away in the month of June to that of July is 4:3."

Which item(s) is the class teacher referring to?

A only mobile **B** only music player
C only mobile and music player **D** only music player and calculator

Skill: Data interpretation/analysis: concept of averages, graph reading, etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
32	28224	B	29%	37%	Your School / All Schools

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 20 A number Δ is represented on the number line shown below.

Which of these numbers is LESS than Δ?

A 2 - Δ **B** 2 + Δ **C** Δ ÷ 2 **D** Δ × 2

Skill: Integers and rational numbers, powers and bases

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
20	28224	D	62%	40%	Your School	
					All Schools	

Q 27 The average temperature on the first three days of a week was 5°C.

If the average temperature on the first two days was 9°C, what was the temperature on the third day?

A 7°C **B** 3°C **C** -3°C **D** -7°C

Skill: Data interpretation/analysis: concept of averages, graph reading, etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
27	28224	C	50%	29%	Your School	
					All Schools	

Q 8 When the operation @ is applied on two numbers, it gives the result as shown here.

✍ **8 @ 9 = 80**
✍ **11 @ 7 = 88**
✍ **24 @ 16 = 408**
✍ **300 @ 30 = 9300**
✍ **933 @ 478 = M**

It is known that 933 × 478 = 445974. What will be the number that will come in the place marked ' **M** '?

A 1 less than 445974 **B** 8 less than 445974
C 478 more than 445974 **D** 933 more than 445974

Skill: Number sense, related competency and computation skills

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
8	28224	D	74%	53%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

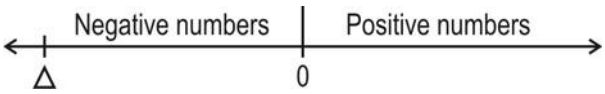
Q 22 p and t are two integers such that $(-p) \times (-t) < 0$.
Which of the following is true?

A both $p > 0$ and $t > 0$
B either $p < 0$ or $t < 0$
C both $p < 0$ and $t < 0$
D (Product of $(-p)$ and $(-t)$ can never be less than zero.)

The question tests understanding of integer multiplication. Students selecting option B have correctly understood that either (-p) or (-t) should be negative for the product to be negative. Students selecting option D thinks that (-p) and (-t) are always negative and hence the product can never be negative. Students selecting option C may have believed that as the expression in the question is less than 0, both p and t should be less than zero. Students selecting option A may have believed that multiplication of negative numbers is also negative. Explain that the expression (âˆ'p) represents the opposite of p, so if p is negative, (-p) becomes positive. Use examples where p is both positive and negative to illustrate this concept. Reinforce understanding by having students practice determining the sign of (âˆ'p) for various values of p.

Papercode: 28224, Question no: 22					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	3%	35%	38%	24%	34
National	12%	27%	23%	38%	12084

Q 20 A number Δ is represented on the number line shown below.



Which of these numbers is LESS than Δ?

A 2 - Δ **B** 2 + Δ **C** Δ ÷ 2 **D** Δ × 2

This question tests if the students have understood the operations using negative numbers. Students selecting option D correctly understood that twice the value of a negative number is less than that number. Students selecting option A thinks that subtraction results in a number less than the two numbers. They have not understood that subtracting a negative number from 2 results in a positive number. Students selecting option B and option C are randomly selecting these options. They may not have understood that addition and division by a positive number result in a number greater than delta.

Papercode: 28224, Question no: 20					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	29%	3%	6%	62%	34
National	25%	17%	16%	40%	12084

Q 35 Which of the following numbers lies between (-11) and (-10)?

A $-\frac{11}{10}$ **B** $-\frac{10}{11}$ **C** -10.11 **D** -11.10

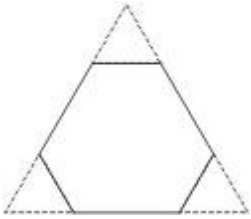
This question tests the understanding of the concept of rational numbers between two numbers. Option C is the correct answer since -10.11 is less than -10 and greater than -11. Option A and B shows an incorrect understanding that a/b or b/a lies between a and b. Option D shows an incorrect understanding of integers.

Papercode: 28224, Question no: 35					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	15%	26%	53%	6%	34
National	21%	24%	44%	9%	12084

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Mensuration: area and perimeter; volume (43%)

Q 1 Rudraksh cuts off three equilateral triangles, of sides 2 cm each, from the three corners of a bigger equilateral triangle.

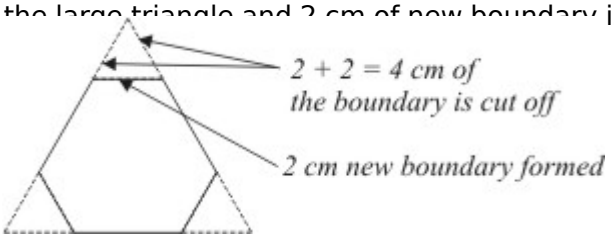


What will be the perimeter of the new shape as compared to the original triangle?

A It will be 6 cm more.
B It will be 6 cm less.
C It will be 18 cm less.
D (We can't say unless the sides of the larger triangle are known.)

Explanation:

B: When one small equilateral triangle is cut off, we see that 4 cm is cut off from the boundary of the large triangle and 2 cm of new boundary is formed.



The perimeter of the bigger triangle decreases by 2 cm, when one small triangle is cut off. So it will decrease by 6 cm when three such triangles are cut off. Note that we just need to calculate the change in perimeter of the larger triangle and not its actual perimeter. Change in perimeter can be calculated from the dimensions of the smaller triangle cut off.

Q 2 A single straw is bent to form various shapes, one after the other, as shown below.

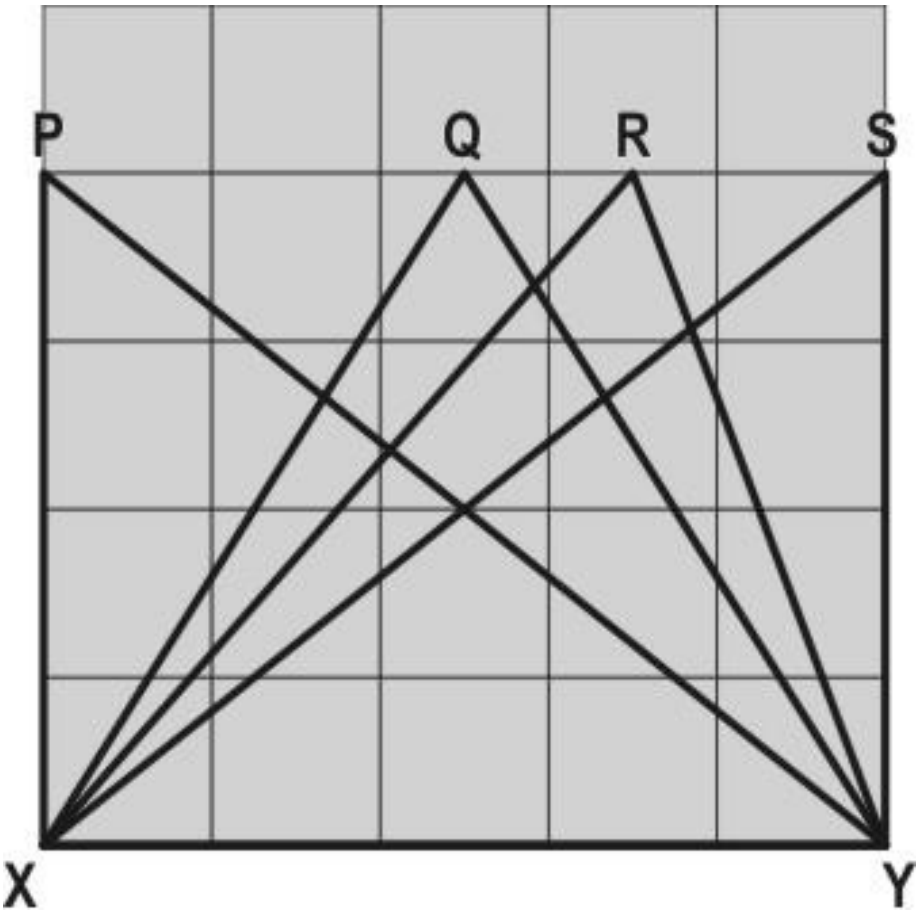
Which of them will have the maximum perimeter?

A A triangle, because each of its sides is bigger than a single side of a square or a hexagon.
B A square, because of all the shapes, a square always has the maximum perimeter.
C A hexagon, because it has more sides than the other two shapes.
D All the shapes will have the same perimeter.

Explanation:

D: The perimeter of a figure is the length of its boundary. Here, the same straw is bent to form various shapes. How will you find the length of the boundary of these shapes? Whatever shape you make with the straw, the straw forms the boundary of the shape. So the length of the boundary, or the perimeter of all these shapes will be equal to the length of the straw. Hence the perimeters of all these shapes are equal. Though the hexagon has a larger number of sides than say the triangle, can you see that each side of the hexagon is smaller than each side of the triangle?

Q 3 Shown here are four triangles - △XPY, △XQY, △XRY and △XSY.



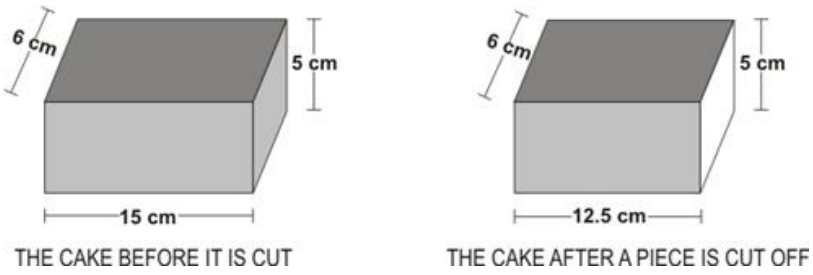
Which of these is true about the four triangles?

A All of them have the same area.
B All of them have different perimeters.
C Only △XPY and △XSY have same area.
D Only △XQY and △XRY have same perimeter.

Explanation:

A: How do we calculate the area of a triangle? $\frac{1}{2}$ × base × height. Here for all the 4 triangles shown, the base is the same namely XY. Can you also see that the heights of all four triangles are equal, being equal to 4 units, where 1 unit is the side of the small square? So the heights and bases of all 4 triangles are of equal length, and hence they are of the same area. Though triangles XQY and XRY look different from triangles XPY and XQY, they all enclose the same area which can be calculated as $\frac{1}{2}$ × length of XY × 4 = twice length of XY.

Q 4 Salim bought a fruitcake, cut off a piece from it, and ate the piece. The dimensions of the cake before and after he cut it, are shown below.



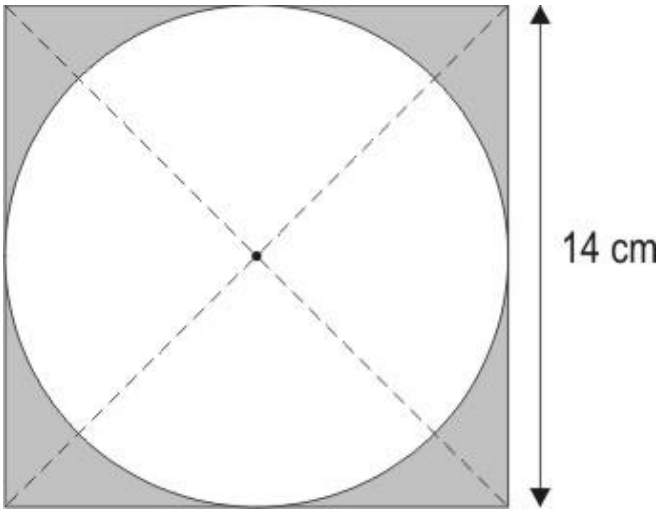
What was the volume of the piece that Salim ate?

A 2.5 cu cm **B** 13.5 cu cm **C** 75 cu cm **D** 375 cu cm

Explanation:

C: The length, breadth and height of the fruit cake were 15 cm, 6 cm and 5 cm respectively. After Salim ate a piece off it, we see that the breadth and height are still 6 cm and 5 cm respectively, but the length has decreased to 12.5 cm. From this we understand that Salim has cut the piece along the length, and had a piece 2.5 cm (15 - 12.5) long. The breadth and width of the piece are the same as the original piece, 6 cm and 5 cm. So its volume is $2.5 \times 6 \times 5 = 75$ cm^3. Note that we are asked for the volume of the piece that Salim ate and not it's length.

Q 5 A circle is inscribed in a square of side 14 cm as shown here.



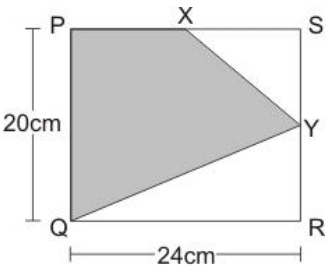
What is the area of the region shaded? *(Note: Use $\pi = \frac{22}{7}$.)*

A 42 cm^2
B 112 cm^2
C 196 cm^2
D (cannot be determined)

Explanation:

A: The shaded region is what would remain if we were to cut out the circle from the square. The area of the square = $14 \times 14 = 196$ cm^2. The diameter of the circle is same as the side of the square, 14 cm. So its radius is 7 cm. The area of the circle = πr^2 (where r is the radius) = $\frac{22}{7} \times 7 \times 7 = 154$ cm^2. So the area of the shaded portion is $196 - 154 = 42$ cm^2. Though you cannot apply any standard formula to find the area of the shaded region, you must see if you can use known areas in a meaningful way and find the required area.

Q 6 The sides of rectangle PQRS are 24 cm and 20 cm. X and Y are the midpoints of sides PS and SR.



What is the area of quadrilateral PXYQ?

A 180 sq cm
B 240 sq cm
C 300 sq cm
D 320 sq cm

Explanation:

C: The area of the quadrilateral PXYQ is obtained by subtracting the area of triangles YSX and YRQ from the area of the rectangle PQRS. Since X and Y are midpoints of the sides PS and SR, XS = $\frac{1}{2}$ PS = 12 cm. SY = YR = $\frac{1}{2}$ SR = 10 cm.

Area of triangle YSX = $\frac{1}{2} \times$ XS $\times$ YS = $\frac{1}{2} \times 12 \times 10 = 60$ sq. cm

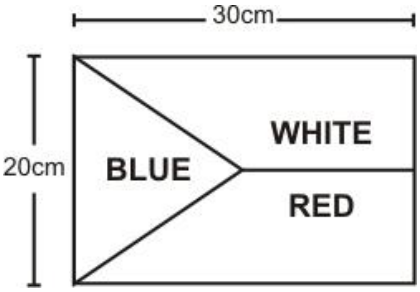
Area of triangle YRS = $\frac{1}{2} \times$ YR $\times$ QR = $\frac{1}{2} \times 24 \times 10 = 120$ sq. cm

Area of rectangle PQRS = PS $\times$ SR = $20 \times 24 = 480$ sq. cm

Therefore area(PXYQ) = area(PQRS) - area(YSX) - area(YRS)= 480 - 60 - 120 = 300 sq.cm.

Note that PXTQ is not a triangle and its area cannot be calculated using the formula $\frac{1}{2} \times$ base $\times$ height.

Q 7 A group of students from the Czech Republic are going to visit Farhan's school. Farhan is asked to paint a Czech flag to put up in his classroom.
The dimensions and colours of the flag are as shown:



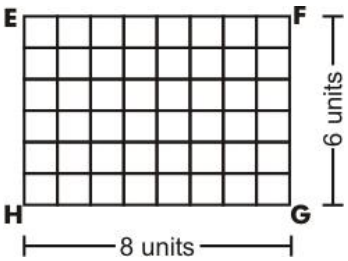
If one vertex of the triangle is exactly at the centre of the flag, how much area would Farhan paint blue?

A 150 sq.cm **B** 200 sq.cm **C** 300 sq.cm **D** 400 sq.cm

Explanation:

A: The vertex of the triangle is at the centre of the flag. So the altitude or height of the triangle will be half the length of the flag that is 15 cm. The base of the triangle is 20 cm. Therefore the area of the triangle is $\frac{1}{2} \times$ base $\times$ altitude = $\frac{1}{2} \times 20 \times 15 = 150$ sq. cm. Note that the altitude of the triangle is only *half* the length of the flag and not the entire length.

Q 8 A rectangle EFGH is shown below:

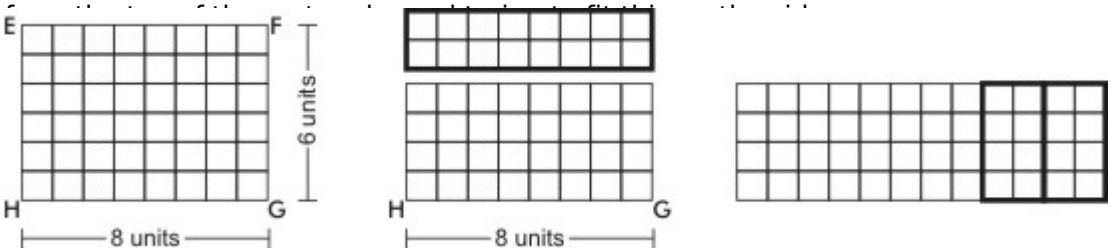


If the width of rectangle EFGH is decreased by 2 units, by how many units should the length be increased so that the AREA of the rectangle remains unchanged?

A 2 **B** 4 **C** 8 **D** 16

Explanation:

B: The area of the rectangle is 8 × 6 = 48 units. The width of the rectangle is decreased by 2 units. That is, it becomes 4 units. Now what should be the length of a rectangle whose area is 48 square units and width 4 units? 48 ÷ 4 = 12 units. So the length should be increased by 4 units from 8 units to 12 units. You could also imagine cutting away a strip 2 units wide and 8 units long



Are you able to see that this increases the width by 4 units?

Skill: Algebra: concepts and applications (43%)

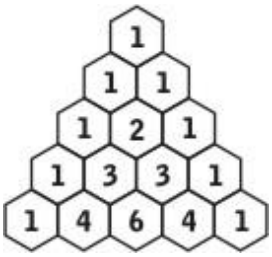
Q 1 If '2 n ' is the general form of writing any even integer, which of the following will NECESSARILY be an odd integer?

A n - 1 **B** 2 n - 1 **C** 2(n + 1) **D** 2 n + 2

Explanation:

B: 2 n is an even integer for any integer n . 2 n - 1 is one less than 2 n and hence will be an odd integer. What can you say about 2 n + 1? It is one more than 2 n and hence an odd integer, as well. But look at 2(n + 1). This is twice (n + 1), and therefore has 2 as a factor. So it is even. We may also say 2(n + 1) = 2 n + 2; ie, 2 more than 2 n , and hence its value is an even integer.

Q 2 See this number pattern carefully. Every row of numbers has a certain relation with the numbers in the row just above it.



Given below is a part of this pattern.

What number should replace '***x***' ?

A 7.5 **B** 5 **C** 4 **D** 3

Explanation:

B: See how the pattern is constructed. How would you write the next row? The first number is of course 1. The second number would be the sum of the first two numbers of the previous row, (1 + 4 = 5), the third number would be the sum of the second and third numbers of the previous row (4 + 6 = 10) and so on. Are you able to see that the number in each hexagon is the sum of the numbers in the two hexagons right above it? So in the given pattern, $x + 2x = 15$ ie $3x = 15$, $x = 5$.

Q 3 A new operation

★

is defined as follows:

p

★

q=2q - p for all rational numbers p and q.
According to this, what will be the value of 5

★

1?

A -8 **B** -3 **C** 8 **D** 9

Explanation:

B: Given that p★ q = 2q - p. So 5★ 1 = 2 × 1 - 5 = -3. Note that the rule is twice the second number (ie q) - the first number(ie p) So 5★ 1 = 2 × 1 - 5 and *not* 2 × 5 - 1 = 9.

Q 4 ♣ and ♦ are two numbers and ♣ + ♦ = 15.

The value of which of the following can be determined from this?

A ♣ − ♦ **B** ♦ × ♣

C 5♦ + 5♣ **D** 5♣ - 5♦

Explanation:

C:

The sum, ♣ + ♦ is 15. Now 5 ♦ + 5 ♣ = 5 ♣ + 5 ♦ = 5(♣ + ♦) = 5 × 15 = 75. 5 ♦ + 5 ♣ is nothing but 5 times the sum ♣ + ♦, and can be determined as we know this sum. Though we know the sum ♣ + ♦, we do not know the individual values of ♣ and ♦. The values could be 11 and 4, or 10 and 5 or even 14.5 and 0.5.Hence cannot find out the values of ♦ × ♣ or ♣ - ♦ from the given information alone.

Q 5 In a 2-digit number, *a* and b are the digits at its tens and ones place respectively.

Which of the following is the 2-digit number?

A $a \times b$ **B** $10a + b$
C $10b + a$ **D** $10(a + b)$

Explanation:

B: 74 is a two digit number with 7 at the tens place and 4 at the ones place. $74 = 10 \times 7 + 4$. We all know that 74 is not the same as 7×4. Similarly if we have a two digit number with a at the ten's place and b at the ones place, the value of the number would be $10 \times a + b = 10a + b$.

Q 6 The table below shows a relation between x and y:

x	1	2	3	4
y	1	5	9	13

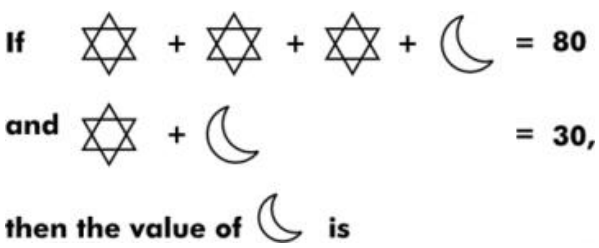
For what value of x will y equal 37?

A 8 ½ **B** 10 **C** 40 **D** 145

Explanation:

B: Study the table carefully. Are you able to figure out the relationship between x and y ? Remember that this should be satisfied by ALL pairs of x and y given. For example for the pair of values 3, 9, y is three times x . But this relation does not hold for all pairs of values. Here the y values are 3 less than four times the x values. ($y = 4x - 3$) So to find 4 times the x value, we add 3 to the y value. If the y value is 37, 4 times the x value is 40 (and not 37 - 3 = 34) Now the x value is $40 \div 4 = 10$.

Q 7

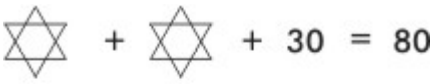


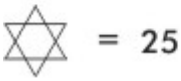
A 5 **B** 10 **C** 15 **D** 25

Explanation:

A: We are given that

. Substituting the value in the first equation and simplifying we get





So we see that star has the value 25. Using this in of the given equations we see that

= 5

Q 8 The formula for finding the perimeter (P) of a rectangle when the length (l) and breadth (b) are given is:
P = 2 (l + b)
Which of the following is the correct sequence of steps to obtain the length of a rectangle when the perimeter and breadth are known?

A Subtract b from P and then divide the result by 2.
B Divide P by b and then divide the result by 2.
C Multiply P by 2 and then add b to the result.
D Divide P by 2 and then subtract b from the result.

Explanation:

D: Perimeter P is twice the sum of the length and and breadth of the rectangle. Dividing P by 2 gives the sum of the length and breadth. Now subtracting the breadth from this gives the length. Note that P is twice the length + twice the breadth. (2 *l* + 2 *b*) When we subtract *b* from P we are left with twice the length + the breadth. ((2 *l* + 2 *b)* - *b* = 2 *l* + *b*) Dividing this by 2 will not give us the length.